

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse, across academia, public policy, business, and popular media, spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

There have been rapid improvements in AI capabilities in several areas. For instance, according to the most recent AI Index Report, AI systems could solve just 4.4% of coding problems on SWE-Bench, a widely used benchmark for software engineering, in 2023, but performance increased to 71.7% in 2024 (Maslej et al., 2025).¹ AI has improved on other benchmarks as well including language understanding, subject knowledge, and reasoning. At the same time, AI systems are increasingly widely adopted. According to Hartley et al. (2025), LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.²

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.³

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through July 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high degree of

¹SWE-bench is designed to evaluate the performance of large language models (LLMs) on real-world software engineering tasks. It uses a collection of GitHub issues to assess an LLM’s ability to generate code that resolves those issues.

²Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% of employed respondents saying they had used generative AI for work at least once in the previous week, and 9% every work day.

³Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

Quarterly Census of Employment and Wages (QCEW).¹⁰ These prior papers use data that lack either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹¹ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this information to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several sample restrictions for our main analysis sample. We include only workers employed by firms that use ADP’s payroll product to maintain worker earnings records. We also exclude employees classified by firms as part-time from the analysis and subset to people between the age of 18 and 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms across our main sample period by keeping only companies that have employee earnings records for each month from January 2021 through July 2025.

In addition, ADP observes job titles for about 70% of workers in its system. We exclude workers who do not have a recorded job title. There are over 7,000 standardized job titles, examples of which include “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of

¹⁰Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹¹As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures described below in Section 3.2.

After these restrictions we have records on between 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two different approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#). [Eloundou et al. \(2024\)](#) estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They then construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second primary approach we take uses data on generative AI usage from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on a sample of several million conversations with Claude, Anthropic’s generative AI model. It then aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it also reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this information as an estimate of whether usage of AI for an occupation is primarily complementary or substitutable with labor.¹⁴

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the Quarterly Census of Employment and Wages (QCEW) using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴Specifically, [Handa et al. \(2025\)](#) first use Claude to classify conversations into six categories: Directive, meaning complete task delegation with minimal interaction; Feedback Loop, meaning task completion guided by environmental feedback such as when repeatedly relaying coding errors to the model; Task Iteration, meaning a collaborative